

DILIPKUMAR NEHRU

Backend Engineer • Node.js • API & Microservices Specialist

Bengaluru, India | dilipkumarnehru75@gmail.com | +91 78456 06064 | linkedin.com/in/dilipkumar-nehru-801050265 | github.com/DilipkumarNehru

Professional Summary

Results-driven Backend Engineer with 4+ years of experience building scalable, high-performance server-side systems using Node.js, Express.js, MongoDB, and MySQL. Delivered production APIs handling 50,000+ daily requests for enterprise CRM and AI analytics platforms. Proven track record in asynchronous processing (RabbitMQ), database performance optimisation, and cross-functional Agile delivery. Complementary QA automation background (Selenium, Postman, JMeter) enables end-to-end ownership from development through release.

Career Highlights

Key quantified achievements across 4+ years:

10,000+ Daily API requests handled on NovoNordisk CRM and UrbanPiper CRM	35% Reduction in API latency via MongoDB indexing	40% Faster regression cycles with Selenium automation
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Work Experience

Backend Engineer – Software Developer | [Triotech Bizserve Private Limited](#) | Bhopal June 2026 – Sep 2026

- Developed and enhanced backend services for Ringingo, a multi-tenant cloud telephony SaaS platform supporting real-time CDR processing, webhook delivery, and API integrations.
- Designed and implemented a RabbitMQ-driven real-time CDR pipeline with timezone-aware scheduled webhooks, incremental cursor-based delivery, and retry/delivery-log mechanisms.
- Built Token-to-API Key management with JWT-based authentication supporting x-api-key and Bearer token mechanisms, including key refresh and expiry handling.
- Delivered end-to-end features across NestJS/Node.js, MongoDB, microservices, and React/Next.js with Tailwind CSS for secure, scalable multi-tenant applications.

Software Engineer – Backend Developer | [Zuqo Corporation India Pvt Ltd](#) | BengaluruDec 2023 – May 2025

- Designed and maintained scalable RESTful APIs using Node.js and Express.js, serving 50,000+ daily requests for enterprise web applications with 99.5% uptime.
- Reduced MongoDB query response times by 35% through strategic indexing, query optimisation, and connection pooling across the NovoNordisk CRM platform.

- ▶ Implemented asynchronous background task processing using RabbitMQ, cutting average job completion time from 8s to under 2s for UrbanPiper AI Analytics.
- ▶ Integrated MongoDB and MySQL databases with backend microservices, standardising schema design patterns that reduced inter-service data inconsistencies by 60%.
- ▶ Collaborated with 6-member cross-functional team in 2-week Agile sprints, consistently delivering features on schedule with zero critical production rollbacks.
- ▶ Debugged and resolved 30+ production incidents, improving application stability and reducing Mean Time to Resolution (MTTR) by approximately 45%.
- ▶ Enforced clean code standards, Git branching strategies (GitFlow), and code review practices, reducing post-merge defect rate by 25%.
- ▶ Implemented JWT-based authentication and OAuth 2.0 authorisation flows across multiple APIs, eliminating session vulnerabilities and enabling secure third-party integrations.
- ▶ Integrated Socket.io to enable real-time WebSocket event notifications and live dashboard updates, reducing client-side polling overhead by 80%.

Software Test Engineer – QA Automation | **Zuqo Corporation India Pvt Ltd** | Bengaluru *Apr 2022 – Dec 2023*

- ▶ Built and maintained a Selenium automation test suite covering 200+ functional and regression test cases, cutting manual testing effort by 40%.
- ▶ Performed comprehensive API testing and backend validation using Postman and JMeter, identifying 80+ defects before production deployment.
- ▶ Created structured bug reports, test plans, and troubleshooting runbooks adopted as the QA documentation standard across the team.
- ▶ Collaborated directly with backend developers to triage and resolve defects, reducing average bug resolution cycle from 5 days to 2 days.

Project Highlights

NovoNordisk CRM — Healthcare Enterprise Platform

- ▶ Architected and delivered scalable backend APIs (Node.js, Express.js, MongoDB) handling 50,000+ daily requests for a global pharmaceutical CRM system.
- ▶ Implemented JWT authentication, OAuth 2.0 flows, field-level encryption, and RBAC for HIPAA-aligned secure data handling of sensitive patient and sales records.
- ▶ Optimised 15 high-frequency MongoDB aggregation queries, reducing dashboard load times by 35% and improving UX for 500+ concurrent users.

UrbanPiper AI Analytics — F&B Analytics Dashboard

- ▶ Built event-driven backend microservices with RabbitMQ and Socket.io WebSockets to process and push real-time analytics data streams to 100+ restaurant client dashboards.
- ▶ Reduced asynchronous job processing latency from 8s to under 2s by redesigning queue consumers and introducing dead-letter queue error handling.
- ▶ Delivered REST APIs consumed by React analytics dashboards, enabling real-time KPI monitoring with sub-500ms response times.

Client Support & Backend Maintenance

- Provided Tier-2 backend production support for 5+ client applications, diagnosing and resolving scalability bottlenecks under SLA constraints.
- Refactored legacy API endpoints, reducing average response time by 20% and eliminating 3 recurring timeout errors in peak-traffic scenarios.

Technical Skills

Languages	JavaScript (ES6+), Node.js, TypeScript (basic), SQL, Python, Java
Frameworks	NestJS, React.js, Next.js, Express.js, Socket.io
Databases	MongoDB, SQL, PostgreSQL
Messaging & Microservices	RabbitMQ, Event-Driven Architecture, WebSockets, CDR Processing
Auth & Security	JWT (JSON Web Tokens), OAuth 2.0, MD5, HTTPS/TLS, bcrypt
Backend	RESTful APIs, WebSockets, Microservices, API Integration, Async Programming, Webhook Integration, Retry Mechanisms
Frontend	React.js, Next.js, Tailwind CSS
Testing	Postman, Selenium WebDriver, JMeter, API Automation, Manual Testing
DevOps / Tools	Git, Bitbucket, GitHub, Linux CLI(working knowledge)
Methodologies	Agile Methodology, CI/CD awareness

Certifications

- Selenium Automation & Manual Testing — QSpiders, Bangalore (2022)
- MongoDB Basics (M001) — MongoDB University [Recommended Next]
- Node.js Application Developer — OpenJS Foundation [Recommended next]

Education

Bachelor of Engineering – Electronics & Communication Engineering (ECE) *Graduation: 2020*

RVS Technical Campus, Coimbatore (Anna University)

Diploma – Electronics & Communication Engineering (ECE)

2012 – 2015

Krishnasamy Memorial Polytechnic College, Cuddalore